

Bluestock Mutual Fund Analytics Capstone Project

Project Documentation

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1. Introduction

A mutual fund is a professionally managed investment vehicle that pools money from multiple investors and invests it in a diversified portfolio of financial assets such as stocks, bonds, money market instruments, and other securities. Mutual funds provide investors with an opportunity to participate in the financial markets without requiring extensive investment knowledge. They are managed by experienced fund managers who make investment decisions based on market conditions and predefined investment objectives. Due to their diversification, professional management, liquidity, and affordability, mutual funds have become one of the most preferred investment options for both individual and institutional investors.

Over the past few years, the mutual fund industry has experienced significant growth due to increasing financial awareness, digital investment platforms, and the availability of Systematic Investment Plans (SIPs). Investors today have access to thousands of mutual fund schemes across different categories, including equity funds, debt funds, hybrid funds, index funds, sectoral funds, and exchange-traded funds. Each scheme differs in terms of risk, return, investment objective, and market performance, making the selection process increasingly complex.

Evaluating the performance of mutual funds requires analyzing several financial indicators rather than relying solely on returns. Metrics such as Net Asset Value (NAV), Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Assets Under Management (AUM), and investor inflows provide valuable insights into both profitability and investment risk. These indicators help investors understand the consistency, volatility, and efficiency of different mutual fund schemes, enabling more informed investment decisions.

With the rapid growth of financial data, manual analysis has become increasingly difficult and time-consuming. Financial institutions and investors require efficient analytical techniques to process large datasets, identify trends, compare fund performance, and generate meaningful insights. Modern data analytics techniques enable faster processing of financial information while improving the accuracy and reliability of investment analysis.

This project applies data analytics techniques to mutual fund data collected from multiple financial sources. The collected data is cleaned, processed, stored,

and analyzed using Python, SQL, and SQLite. Various financial performance metrics are calculated to evaluate the performance and risk associated with different mutual fund schemes. The project aims to provide a structured approach for analyzing mutual fund performance and generating useful insights that can support investors, financial analysts, and researchers in understanding market trends and evaluating investment opportunities more effectively.

2. Objectives

The primary objectives of the **Mutual Fund Analytics Dashboard** project are:

- To collect mutual fund datasets from reliable financial sources.
- To clean, preprocess, and validate the raw data for accurate analysis.
- To store the processed data efficiently using an SQLite database.
- To perform Exploratory Data Analysis (EDA) to identify trends, patterns, and relationships in the data.
- To calculate important financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- To analyze mutual fund performance based on returns, risk, AUM, investor inflows, and other financial indicators.
- To develop an automated data analytics pipeline using Python for efficient data processing.
- To write SQL queries for effective data retrieval and management.
- To create interactive and user-friendly dashboards using Microsoft Power BI for data visualization.
- To compare different mutual fund schemes using performance and risk metrics.
- To generate fund recommendations based on calculated financial metrics.
- To provide meaningful insights that support investors and financial analysts in making informed investment decisions.

3. Problem Statement

The rapid growth of the mutual fund industry has resulted in the availability of a large number of investment schemes across various categories, making it increasingly difficult for investors to identify suitable investment options. Each mutual fund differs in terms of returns, risk level, asset allocation, fund size, and market performance. Evaluating these schemes requires analyzing multiple financial indicators such as Net Asset Value (NAV), Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Assets Under Management (AUM), and investor inflows. Performing such

analysis manually is both time-consuming and prone to errors, particularly when dealing with large volumes of historical financial data.

Most investors rely on scattered financial reports, websites, or individual performance metrics that do not provide a comprehensive view of mutual fund performance. This fragmented approach makes it difficult to compare funds effectively and understand the relationship between returns and associated risks. As a result, investors may make decisions based on incomplete or inconsistent information.

Additionally, financial datasets often contain missing values, duplicate records, and inconsistent formats, making data preprocessing an essential step before meaningful analysis can be performed. Without a structured data analytics pipeline, extracting useful insights from mutual fund data becomes challenging.

Therefore, there is a need for an automated system that can collect, clean, process, analyze, and organize mutual fund data efficiently. Such a system should compute important financial performance metrics, perform analytical comparisons, and present meaningful insights in a clear and interactive manner. By automating these processes, investors and financial analysts can save time, reduce manual effort, improve analytical accuracy, and make more informed investment decisions based on reliable data.

4. Technologies Used

Technology	Purpose
Python	Data Processing
Pandas	Data Manipulation
NumPy	Numerical Computation
SQLite	Database
SQL	Querying Data
Jupyter Notebook	Analysis
Power BI	Dashboard
Matplotlib	Data Visualization
Requests API	Live NAV Data

5. Dataset Description

The Mutual Fund Analytics project utilizes multiple datasets that provide comprehensive information about mutual fund schemes, investor activities, market benchmarks, and fund performance. These datasets were collected from publicly available financial sources and processed to support data analysis, performance evaluation, and visualization.

The datasets used in this project are described below:

- **Fund Master**

Contains the basic information of mutual fund schemes, including fund names, AMFI codes, categories, fund houses, and scheme details. This dataset serves as the master reference for all mutual fund records.

- **NAV History**

Includes historical Net Asset Value (NAV) data of mutual fund schemes. It is used to analyze fund performance over time and calculate returns.

- **AUM by Fund House**

Provides information about the Assets Under Management (AUM) for different fund houses. It helps compare the market size and popularity of various asset management companies.

- **Monthly SIP Inflows**

Contains monthly Systematic Investment Plan (SIP) investment data. This dataset is useful for analyzing investor participation and investment trends over different periods.

- **Category Inflows**

Represents net inflows and outflows across different mutual fund categories such as Equity, Debt, Hybrid, and Index Funds. It helps identify investor preferences.

- **Industry Folio Count**

Provides the number of investor folios across the mutual fund industry. This dataset indicates investor growth and market participation.

- **Scheme Performance**

Contains historical performance information of mutual fund schemes, including returns over different time horizons. It is used to evaluate and compare fund performance.

- **Investor Transactions**

Includes details of investment transactions such as purchases, redemptions, and switches. It supports the analysis of investor behavior and transaction patterns.

- **Portfolio Holdings**

Provides information about the securities held by mutual fund schemes, including equity shares, bonds, and other financial instruments. It helps analyze portfolio diversification.

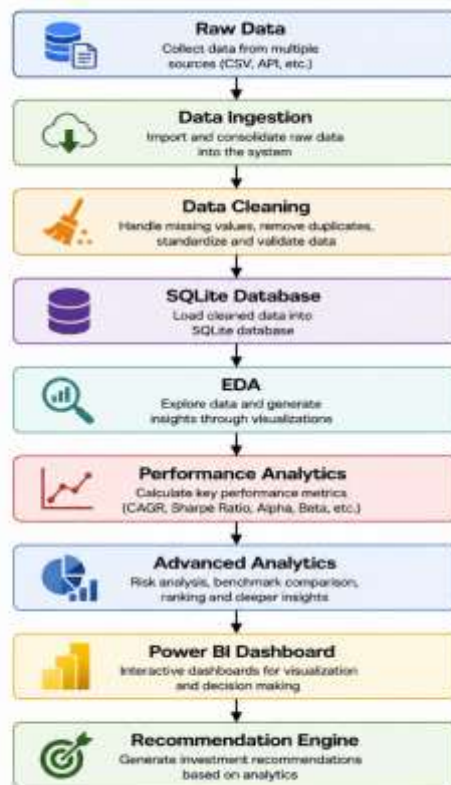
- **Benchmark Indices**

Contains historical data of benchmark market indices such as Nifty 50 and Nifty 100. These indices are used to compare mutual fund performance against the overall market.

- **Live NAV Data**

Includes the latest Net Asset Value (NAV) obtained through an API. This dataset enables the project to analyze recent market performance and maintain updated fund information.

6. Project Architecture



7. Project Workflow

The proposed Mutual Fund Analytics Dashboard follows a structured data processing pipeline that transforms raw financial data into meaningful insights and investment recommendations. Each stage of the workflow performs a specific task to ensure data accuracy, efficient analysis, and interactive visualization.

1. Raw Data Collection

The project begins by collecting mutual fund data from multiple sources, including CSV files, APIs, and publicly available financial datasets. These datasets contain information related to mutual fund schemes, NAV history, AUM, SIP inflows, benchmark indices, investor transactions, and portfolio holdings.

2. Data Ingestion

In this stage, all raw datasets are imported into the project using Python. The collected data from different sources is consolidated into a common format to facilitate further processing and analysis.

3. Data Cleaning

The imported datasets are preprocessed to improve data quality. Missing values, duplicate records, and inconsistent formats are handled appropriately. Data validation and standardization ensure that the datasets are reliable and suitable for analysis.

4. SQLite Database

After cleaning, the processed datasets are stored in an SQLite database. This centralized database enables efficient data storage, retrieval, and SQL-based querying, forming the foundation for further analytical tasks.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand the characteristics of the datasets. Various statistical summaries and visualizations are generated to identify trends, distributions, relationships, and anomalies within the mutual fund data.

6. Performance Analytics

Key financial performance metrics are calculated to evaluate mutual fund performance. These metrics include Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. These indicators help assess both returns and associated investment risks.

7. Advanced Analytics

Advanced analytical techniques are applied to compare mutual fund performance, evaluate benchmark indices, rank funds based on multiple financial metrics, and generate deeper insights into investment behavior and market trends.

8. Power BI Dashboard

The processed data and calculated performance metrics are integrated into Microsoft Power BI to create interactive dashboards. These dashboards provide visual representations of fund performance, risk analysis, SIP trends, benchmark comparisons, and investor insights, enabling users to explore the data effectively.

9. Recommendation Engine

Based on the calculated financial metrics and analytical results, the recommendation engine identifies suitable mutual fund schemes. It generates investment recommendations by considering factors such as returns, risk levels, and overall fund performance, thereby supporting informed investment decisions.

8. Data Ingestion

Data ingestion is the process of collecting and importing raw data from multiple sources into the analytics system for further processing. In this project, data ingestion serves as the initial step in the data pipeline, where mutual fund datasets are gathered from publicly available CSV files and live financial APIs. The collected datasets include Fund Master, NAV History, AUM by Fund House, Monthly SIP Inflows, Category Inflows, Industry Folio Count, Scheme Performance, Investor Transactions, Portfolio Holdings, Benchmark Indices, and Live NAV data.

Python was used to automate the data ingestion process. Libraries such as Pandas were used to read CSV files, while the Requests library was used to fetch live NAV data from APIs. During ingestion, the datasets were validated to ensure successful loading and correct formatting. The imported data was then organized into the project's raw data directory, making it ready for preprocessing and further analysis.

The automated data ingestion process reduces manual effort, improves data consistency, and enables the project to efficiently handle multiple datasets. It also provides a scalable foundation for future updates, allowing new mutual fund data to be incorporated into the system with minimal modifications.

9. Data Cleaning

The following data cleaning steps were performed to improve the quality and consistency of the mutual fund datasets:

- Imported the raw datasets into Python using the Pandas library.
- Identified and handled missing (null) values in the datasets.
- Removed duplicate records to avoid redundant data.
- Corrected inconsistent data formats and standardized column names.
- Converted date columns into a uniform datetime format.
- Converted numerical columns to appropriate data types for analysis.
- Removed unnecessary or irrelevant columns from the datasets.

- Checked for invalid and inconsistent data entries and corrected them wherever possible.
- Standardized categorical values to maintain consistency across datasets.
- Verified data integrity after cleaning to ensure accuracy.
- Saved the cleaned datasets into the **data/processed** folder for further analysis.
- Prepared the cleaned data for loading into the SQLite database and subsequent analytical processes.

10. Database Design

The project uses **SQLite** as the database management system to store and manage the cleaned mutual fund datasets. SQLite is a lightweight, serverless, and file-based relational database that is easy to integrate with Python applications. It provides efficient storage, fast data retrieval, and supports SQL queries for data analysis.

After completing the data cleaning process, the processed datasets are loaded into the SQLite database. Organizing the data into structured tables improves data consistency, minimizes redundancy, and enables efficient querying during analytical operations and dashboard development.

The database serves as the central repository for all processed mutual fund data and supports the generation of financial metrics, analytical reports, and Power BI visualizations.

Database Tables

The SQLite database consists of the following major tables:

- **Fund Master** – Stores basic information about mutual fund schemes, including AMFI code, scheme name, category, and fund house.
- **NAV History** – Contains historical Net Asset Value (NAV) records used for return calculations and trend analysis.
- **Scheme Performance** – Stores historical performance details of mutual fund schemes, including return-related information.
- **Investor Transactions** – Maintains records of investor activities such as purchases, redemptions, and switches.
- **AUM by Fund House** – Stores Assets Under Management (AUM) information for different fund houses.
- **Monthly SIP Inflows** – Contains monthly SIP investment data used for investor trend analysis.

- **Category Inflows** – Stores category-wise inflow and outflow information.
- **Industry Folio Count** – Maintains the number of investor folios across the mutual fund industry.
- **Portfolio Holdings** – Contains details of securities held by mutual fund schemes.
- **Benchmark Indices** – Stores benchmark market index data such as Nifty 50 and Nifty 100 for performance comparison.

SQL Scripts

The following SQL scripts were developed as part of the project:

- **schema.sql** – Defines the database schema by creating the required tables and specifying their structure.
- **queries.sql** – Contains SQL queries used for retrieving, filtering, aggregating, and analyzing mutual fund data stored in the SQLite database.

11. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an essential step in the data analytics process that involves examining, summarizing, and visualizing datasets to understand their characteristics before performing advanced analysis. The primary objective of EDA is to identify patterns, trends, relationships, and anomalies within the mutual fund data. It also helps in validating data quality and gaining insights that support further analytical tasks.

In this project, EDA was performed using Python libraries such as Pandas and Matplotlib. Various statistical summaries and graphical visualizations were generated to analyze mutual fund performance, investor behavior, benchmark trends, and risk distribution. These visualizations provide a clear understanding of the data and support the calculation of financial performance metrics.

The following visualizations were generated during EDA:

• NAV Trend

This visualization shows the historical movement of Net Asset Values (NAV) for mutual fund schemes over time. It helps identify growth patterns, market fluctuations, and long-term fund performance.

- **Risk Grade Distribution**

The Risk Grade Distribution chart illustrates the distribution of mutual funds across different risk categories. It enables investors to understand the overall risk profile of available mutual fund schemes.

- **Returns Distribution**

The Returns Distribution graph represents the spread of returns generated by different mutual funds. It helps identify high-performing and low-performing funds and provides an overview of return variability.

- **Top Fund Houses**

This visualization compares leading Asset Management Companies (AMCs) based on selected financial indicators such as Assets Under Management (AUM) or the number of mutual fund schemes. It highlights the major contributors within the mutual fund industry.

- **Transaction Types**

The Transaction Types chart analyzes investor activities such as purchases, redemptions, and switches. It provides insights into investor behavior and transaction patterns across different mutual fund schemes.

- **Benchmark Performance**

The Benchmark Performance visualization compares benchmark market indices such as Nifty 50 and Nifty 100. This comparison helps evaluate whether mutual fund schemes have outperformed or underperformed the market benchmarks.

12. Performance Analytics

In this project, various financial performance indicators were calculated using Python. The computed metrics provide a comprehensive evaluation of mutual fund performance by considering both returns and risk factors. The major financial metrics calculated are described below.

- **Compound Annual Growth Rate (CAGR)**

Compound Annual Growth Rate (CAGR) measures the average annual growth rate of an investment over a specified period, assuming the profits are reinvested each year. It provides a standardized measure for comparing the long-term performance of different mutual funds.

- **Sharpe Ratio**

The Sharpe Ratio measures the risk-adjusted return of a mutual fund by comparing its excess return with the total risk (standard deviation). A higher Sharpe Ratio indicates that the fund generates better returns for the level of risk taken.

- **Sortino Ratio**

The Sortino Ratio is a variation of the Sharpe Ratio that considers only downside risk instead of total risk. It provides a more accurate measure of performance by focusing on harmful volatility, making it useful for evaluating investment quality.

- **Alpha**

Alpha measures the excess return generated by a mutual fund compared to its benchmark index. A positive Alpha indicates that the fund has outperformed the benchmark, while a negative Alpha suggests underperformance.

- **Beta**

Beta measures the sensitivity of a mutual fund's returns relative to market movements. A Beta value greater than one indicates higher volatility than the market, whereas a value less than one indicates lower market risk.

- **Maximum Drawdown**

Maximum Drawdown represents the largest percentage decline in the value of a mutual fund from its highest peak to its lowest point during a specific period. It is an important indicator for assessing potential investment losses and downside risk.

13. Advanced Analytics

In this project, the financial metrics calculated during the performance analytics stage were integrated to generate a unified performance report. The analysis helps investors compare mutual fund schemes, identify high-performing funds, evaluate investment risks, and understand how different funds perform relative to benchmark indices. By combining multiple financial indicators, the project provides a more reliable basis for investment decisions.

The following advanced analytical tasks were performed:

- **Risk-Adjusted Return Comparison**

Mutual funds were compared using risk-adjusted performance measures such as the Sharpe Ratio and Sortino Ratio. This analysis evaluates how efficiently a fund generates returns relative to the level of risk undertaken. Funds with higher risk-adjusted returns are considered more efficient investment options.

- **Top-Performing Funds**

The calculated performance metrics were used to identify the best-performing mutual fund schemes. Factors such as CAGR, Alpha, and overall returns were considered to rank funds according to their long-term performance and consistency.

- **Benchmark Comparison**

The performance of mutual funds was compared against benchmark market indices such as Nifty 50 and Nifty 100. This comparison helps determine whether a fund has outperformed or underperformed the market over a specific period, providing a clearer understanding of fund effectiveness.

- **Investment Ranking**

A ranking system was developed by combining multiple financial indicators, including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. Mutual funds were ranked based on their overall performance, risk profile, and return potential, helping investors identify suitable investment opportunities.

14. Recommendation Engine

The recommendation process combines multiple performance indicators rather than relying on a single metric. This approach provides a balanced evaluation by considering both return potential and investment risk. Based on the calculated financial metrics, each mutual fund is assigned a recommendation level that helps investors make informed investment decisions.

The recommendation categories used in this project are:

- **Highly Recommended** – Funds with excellent long-term returns, strong risk-adjusted performance, and low downside risk.
- **Recommended** – Funds that demonstrate good overall performance with acceptable risk levels.
- **Moderate** – Funds with average performance or moderate investment risk that may be suitable for cautious investors.
- **Avoid** – Funds with relatively poor performance, high risk, or unfavorable financial indicators.

The recommendation engine evaluates mutual funds using the following financial metrics:

- **Compound Annual Growth Rate (CAGR)**

CAGR is used to measure the long-term growth potential of a mutual fund. Funds with consistently higher CAGR values receive better recommendations.

- **Sharpe Ratio**

The Sharpe Ratio measures risk-adjusted returns by comparing excess returns with total investment risk. A higher Sharpe Ratio indicates more efficient performance and contributes positively to the recommendation.

- **Maximum Drawdown**

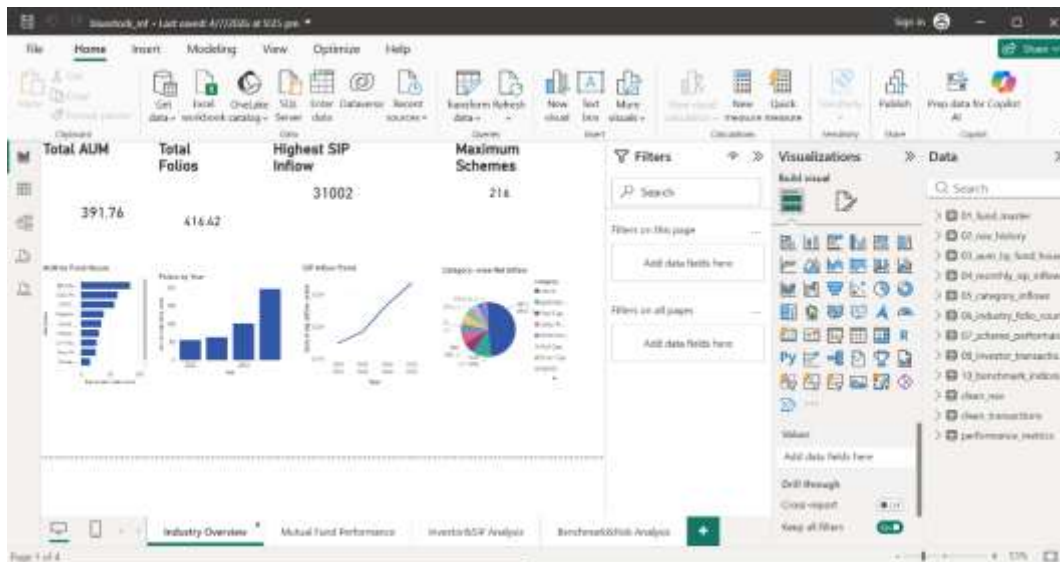
Maximum Drawdown measures the largest decline in a fund's value from its peak. Funds with lower drawdown values are considered less risky and are generally assigned higher recommendation levels.

15. Power BI Dashboard

The dashboard contains four interactive pages:

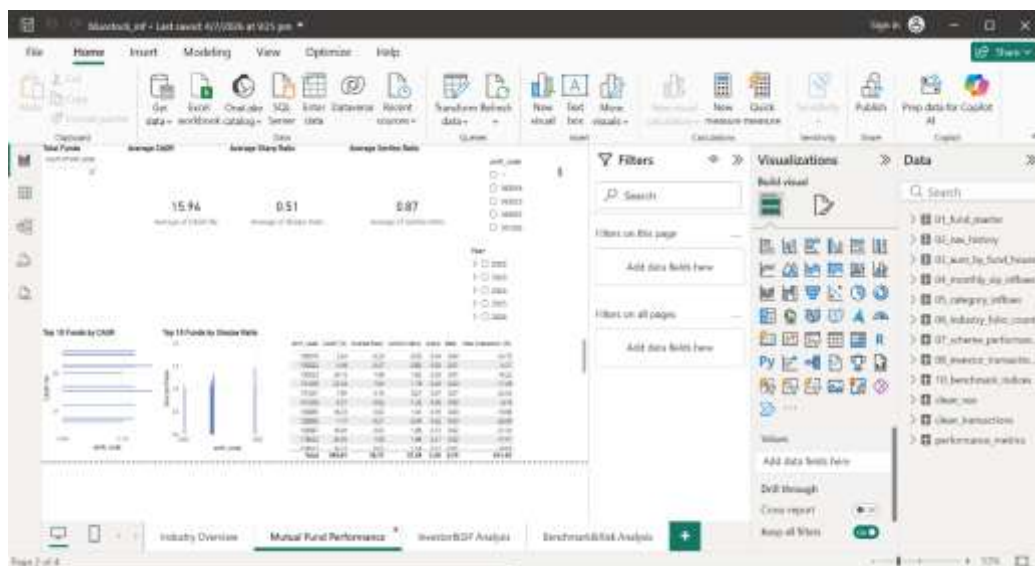
Page 1 – Executive Summary

- Total Funds
- Average Returns
- Total AUM
- NAV Trend



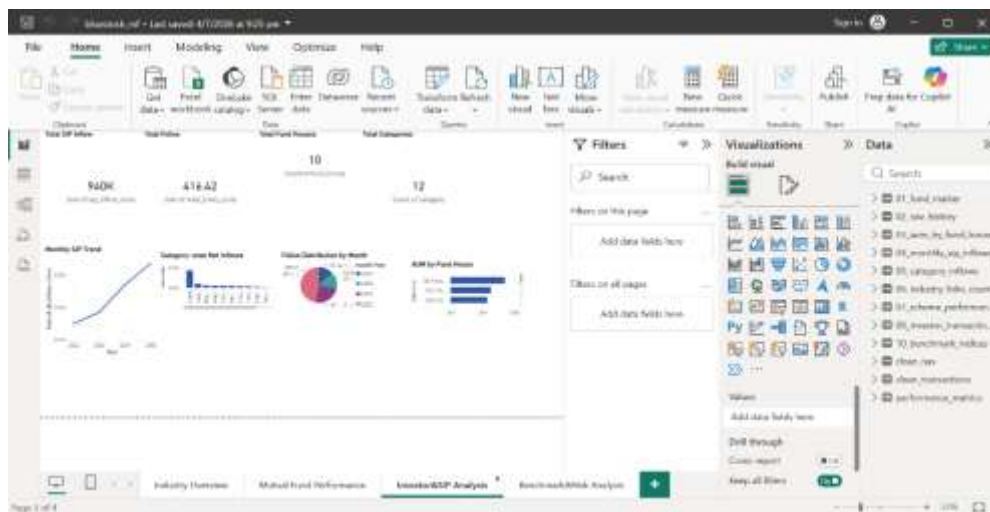
Page 2 – Fund Performance

- CAGR Comparison
- Sharpe Ratio
- Sortino Ratio
- Alpha & Beta



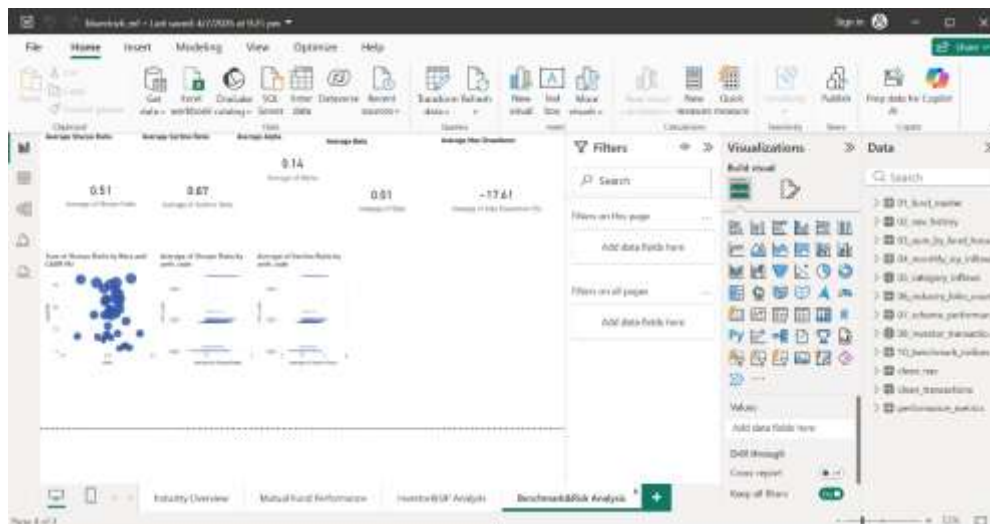
Page 3 – Investor & SIP Analysis

- Monthly SIP
- Category Inflows
- Investor Transactions



Page 4 – Risk Analytics

- Maximum Drawdown
- Benchmark Comparison
- Top Performing Funds
- Risk Distribution



16. Results

The project successfully:

- The developed Mutual Fund Analytics system successfully processed and analyzed multiple mutual fund datasets to generate meaningful insights for investment analysis. The automated ETL pipeline efficiently collected, cleaned, and transformed the raw datasets, which were then stored in an SQLite database for structured data management.
- Financial performance metrics such as Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were successfully calculated for mutual fund schemes. These metrics enabled a comprehensive evaluation of fund performance by considering both returns and associated investment risks.
- The Power BI dashboard provided an interactive visualization platform with multiple analytical pages, including Industry Overview, Mutual Fund Performance, Investor & SIP Analysis, and Benchmark & Risk Analysis. These dashboards allowed users to compare mutual funds, analyze market trends, monitor investor activity, and evaluate benchmark performance through dynamic charts, tables, and key performance indicators.
- Advanced analytics further enhanced the analysis by comparing risk-adjusted returns, identifying top-performing funds, and ranking mutual fund schemes based on multiple financial indicators. Finally, the recommendation engine generated investment recommendations by evaluating the calculated performance metrics, assisting investors in selecting suitable mutual fund schemes according to their performance and risk profile.
- Overall, the project successfully integrated Python, SQLite, SQL, and Power BI into a complete mutual fund analytics solution capable of transforming raw financial data into meaningful insights and supporting data-driven investment decision-making.

File Edit Selection View Go Run

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Save In

DS_advanced_analytics.py x Programiz Python

C:\Users>cmd>Downloads>mutual_fund_analytics>notepad++>DS_advanced_analytics.py>Import pandas as pd

Generate Code Markdown Run All Select Kernel

4 101207 7.938765 0.162661 0.274615 0.069089 0.071078 35.446516

```
summary.sort_values(
    "Sharpe Ratio",
    ascending=False
).head(10)
```

Python

	amfi_code	CAGR (%)	Sharpe Ratio	Sortino Ratio	Alpha	Beta	Max Drawdown (%)
34	148567	30.974108	1.440291	2.368342	0.259250	0.018790	-11.265729
30	120843	30.907455	1.306744	2.352942	0.224506	0.024151	-12.973968
19	119551	25.804666	1.208267	2.103952	0.202612	0.023470	-15.012385
25	120505	32.827406	1.180101	2.010035	0.298101	0.051352	-16.168514
2	100013	30.123153	1.093699	1.823824	0.255114	0.005311	-16.217209
9	118632	24.049493	1.081659	1.839659	0.212215	0.015295	-17.414075
36	148569	27.319588	1.046727	1.802776	0.255039	0.017796	-16.396743
3	101206	23.538361	1.027213	1.785592	0.203189	0.027073	-11.291596
24	120504	23.215119	1.026124	1.785043	0.202956	-0.003047	-12.588276
16	119054	28.214417	0.996231	1.696390	0.233242	0.008052	-20.960884

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Python 3.8.10 (tags/v3.8.10:2c894f958, Oct 14 2020) [AMD64] on win32

17. Conclusion

The Mutual Fund Analytics project was successfully completed by developing a complete data analytics pipeline that transforms raw mutual fund data into meaningful insights. The project began with collecting and processing datasets from multiple sources through an automated ETL pipeline, followed by data cleaning and preprocessing to improve data quality. The processed datasets were then stored in an SQLite database, providing an efficient and organized storage solution for further analysis. Financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to evaluate the return and risk characteristics of various mutual fund schemes. An interactive Power BI dashboard was developed to visualize important trends, comparisons, and performance indicators, enabling users to explore the data effectively. Finally, an investment recommendation engine was implemented to rank mutual funds and generate recommendations based on key financial metrics. Overall, the project demonstrates how data analytics, database management, Python programming, SQL, and Power BI can be integrated to support data-driven investment analysis and assist investors in making informed financial decisions.

18. Future Scope

The current system can be further enhanced with the following improvements:

- **Real-Time Data Integration:** Integrate live mutual fund and market data APIs to provide real-time NAV updates and dashboard refreshes.
- **Machine Learning-Based Recommendations:** Develop an intelligent recommendation system using machine learning algorithms to provide personalized investment suggestions.
- **Portfolio Optimization:** Implement portfolio optimization techniques to help investors build diversified portfolios based on risk and return preferences.
- **Web-Based Application:** Deploy the project as a web application, enabling users to access dashboards and analytics from any device through a browser.
- **Cloud Database and Automation:** Integrate cloud-based databases and automate daily data collection, processing, and report generation for improved scalability and accessibility.

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